Java vs. Python

Java

- Compiled
  - Code is pre-compiled into bytecode
  - Runs in a virtual machine (JVM)
- Static typing
  - Types have to be declared in code
  - Variables cannot change type
  - Not allowed:

    ```java
    int a = 123;
    a = "test";
    ```

- Uses curly brackets to define scope:

  ```java
  public class Summer {
      public int sum(int a, int b) {
          return a+b;
      }
  }
  ```

- Object-oriented by nature
  - All functions are methods
    (with some exceptions)
- Uses packages

Python

- Interpreted
  - Compiled into bytecode at runtime
  - Also runs in a virtual machine
- Dynamic typing
  - Types are not declared
  - Variables can change type during runtime
  - Allowed:

    ```python
    a = 123
    a = "test"
    ```

- Uses whitespace (tabs or spaces) to define scope:

  ```python
  class Summer:
      def sum(self, a, b):
          return a+b
  ```

- Is a multi-paradigm programming language
  - Has both functions and methods
- Uses modules